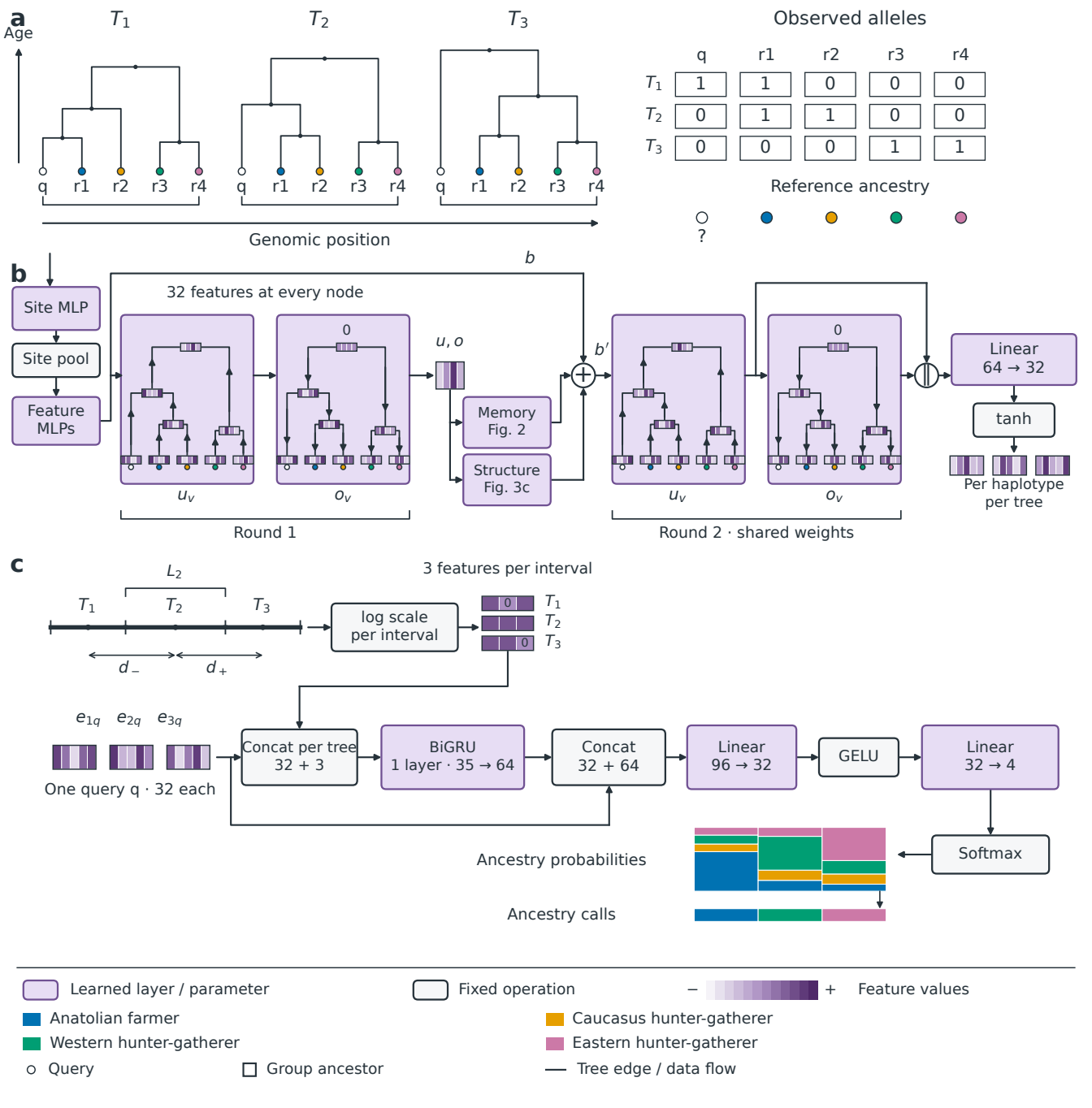
a
Age
T_1
T_2
T_3
q r1 r2 r3 r4
Genomic position
Observed alleles
q r1 r2 r3 r4
T_1 1 1 0 0 0
T_2 0 1 1 0 0
T_3 0 0 0 1 1
Reference ancestry
?
b
Site MLP
Site pool
Feature MLPs
32 features at every node
0
u_v
o_v
u, o
Memory Fig. 2
Structure Fig. 3c
b
b'
Round 1
Round 2 · shared weights
Linear 64 → 32
tanh
Per haplotype per tree
c
L_2
T_1 T_2 T_3
d_- d_+
log scale per interval
3 features per interval
e_{1q} e_{2q} e_{3q}
One query q · 32 each
Concat per tree 32 + 3
BiGRU 1 layer · 35 → 64
Concat 32 + 64
Linear 96 → 32
GELU
Linear 32 → 4
Softmax
Ancestry probabilities
Ancestry calls
Learned layer / parameter
Fixed operation
− + Feature values
Anatolian farmer
Caucasus hunter-gatherer
Western hunter-gatherer
Eastern hunter-gatherer
Query
Group ancestor
Tree edge / data flow